rhythm kept by the baton of a single orchestral conductor: every process dances independently with its neighbours, following its own rhythm. The passing of time is intrinsic to the world, it is born of the world itself, out of the relations between quantum events which *are* the world and which themselves generate their own time.

In fact, the nonexistence of time does not mean anything particularly complicated. Let's try to understand.

The candle chandelier and the pulse

Time appears in most equations of classic physics. It is the variable indicated by the letter *t*. The equations tell us how things change *in time*. If we know what has happened in the past, they allow us to predict the future. More precisely, we measure some variables – for example, the position *A* of an object, the angle *B* of a swinging pendulum, the temperature *C* of an object – and the equations of physics tell us how these variables *A*, *B* and *C* will change with time. They predict the functions *A(t)*, *B(t)*, *C(t)*, and so on, which describe the changing of these variables in time *t*.

Galileo was the first to understand that the movement of objects on Earth could be described by equations for the functions of time *A(t)*, *B(t)*, *C(t)* – and the first to write explicit equations for these functions. The first law of terrestrial physics found by Galileo, for example, describes how an object falls, that is to say, how its altitude *x* varies with the passage of time *t*.[fn40]

To discover and verify this law, Galileo needed two kinds of measurements. He had to measure the height *x* of the object and the time *t*. Therefore, he needed, in particular, an instrument to measure time. He needed a *clock*.

When Galileo lived there were no accurate clocks. Galileo himself, as a young man, discovered a key to making precise timepieces. He discovered that the oscillations of a pendulum all have the same duration (irrespective of the amplitude). Thus, it is possible to measure time by simply counting the oscillations of a pendulum. It seems such an obvious idea, but it took Galileo to find it; it had not occurred to anyone before him. So it goes, with science.

But things are not really this straightforward.

According to legend, Galileo alighted on the idea in Pisa's marvellous cathedral while watching the slow oscillations of a gigantic candle chandelier, which is still there. (The legend is false, since the chandelier was actually first hung there years after Galileo's death, but it makes for a good story. Perhaps there was another one hanging there at the time.) The scientist was observing the oscillations during a religious service in which he was evidently not particularly absorbed, and he was measuring the duration of each oscillation of the chandelier by counting the beats of his own pulse. With mounting excitement, he discovered that the number of beats was the same for each